

AFAMEFUNA UMEJIAKU

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SUMMARY

Experienced ML Engineer specializing in NLP, Computer Vision, Security, model development, and data analytics. Skilled in Python, SQL, and visualization tools, with a track record of building scalable pipelines and delivering actionable insights.

EXPERIENCE

Machine Learning Researcher @ Data Analytics Laboratory | Texas Tech University | Lubbock, TX, USA |

January 2022 - Present

- **Secure Authentication Systems:** Leveraged NLP techniques to enhance the security of authentication mechanisms and preemptively detect system vulnerabilities using OpenAI GPT-4 model, Llama and Langchain.
- **Smart Contract Risk Mitigation:** Applied NLP (Bert) to interpret and validate smart contracts, significantly reducing the probability of financial loss.
- **Network Vulnerability Detection:** Built a network fault detection system using Apache Kafka for real-time data ingestion, AWS Glue for ETL processing, and Snowflake as the centralized data warehouse
- **Additional Projects:** Developed a range of models and applications, including a ChatBot using NLP, fraud detection with XGBoost, exam proctoring via background monitoring in R (dplyr), biometric classroom attendance using CNNs and Raspberry Pi cameras, a secured highway toll system, automated Android testing tool powered by KNN for code review and compliance, buiding scalable data pipelines for sensitive projects. .
- **Publications:** Published 8 peer-reviewed research articles contributing to advancements in cybersecurity and machine learning, with a focus on topics such as password authentication, privacy-preserving AI models, and secure data-driven decision-making frameworks.

Business Development Officer (Information Technology) | University of Jos | Jos, Nigeria | January 2018 - August 2018

- Designed, trained, and deployed robust classification models for recommendation system aimed at streamlining the university admissions process. By automating applicant evaluation and prioritization, the system significantly enhanced operational efficiency and decision accuracy. These improvements led to a 400% surge in admission form purchases and reduced the time from application closure to final admission decisions by 70%, enabling faster onboarding and improved experience for prospective students.
- Built predictive and recommender systems to optimize admission decisions and program targeting to reduce decision timeframe by 60%.
- Performed comprehensive data analyses to assess and forecast the impact of both the consultancy unit's and the university's strategic initiatives.

Head (Information Technology Dept) | Novalson Ventures Nigeria Ltd | Ikeja, Nigeria | August 2014 - September 2017

- Played a key role in securing high-value contracts by modeling a delivery system that reduced delivery time by 30% and cut costs by 40%.
- Trained and deployed predictive models to estimate project timelines, labor needs, and resource allocation, improving planning accuracy using Oracle p6.
- Monitored IT infrastructure around the clock, ensuring swift resolution of outages and minimizing downtime to maintain above 95% system reliability.

EDUCATION

PhD & Msc Computer Science. | Texas Tech University | Lubbock, TX, USA | 2025

Master's, Information Management Technology | Federal University of Technology, Owerri | Owerri, Nigeria | 2016

Bachelor's, Computer Engineering | Nnamdi Azikiwe University, Awka | Awka, Nigeria | 2009

CERTIFICATIONS

AWS Certified Machine Learning – Specialty 2025 | Certified AI Practitioner 2025 | Certified Cloud Practitioner 2025

DeepLearning.AI TensorFlow Developer 2024 | Data Engineering Specialization 2024 | Natural Language Processing 2022

SELECT 3 OF 9 RESEARCH PUBLICATIONS

Enhancing Password Security with Honeywords and Large Language Models (LLMs) Ensemble Summarization Models: Advancing Performance through Collaborative Architectures Balancing Security and Usability: Exploring Prompt-Based Password Generation Models

SKILLS

Languages: Python, JAVA, JavaScript, R, HTML, CSS

Platforms: Git, Docker, AWS, Google Collab, Linux, Kubernetes

Technical Skills: PyTorch, TensorFlow, Stat, Scikit-Learn, Spicy, Pandas, NumPy, Keras, SQL, Terraform, CICD, Matplot-lib